

Joshua Elieson

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Skills

Languages: JavaScript, TypeScript, Java, Python, C/C++, C#, SQL

Technologies & Tools: AWS, Docker, Kubernetes, Linux, Bash, Terraform, React, Next.js, PostgreSQL, Redis, Git, CI/CD, REST APIs, Microservices

Clearance: U.S. Citizen with active U.S. Government Security Clearance (Secret)

Education

Columbia University

Master of Science in Computer Science

Aug 2026 – Dec 2027

Expected Graduation: Dec 2027

University of Utah

Bachelor of Science in Computer Engineering

Aug 2022 – May 2026

GPA: 3.8/4.0

Work Experience

Northrop Grumman

Aug 2024 – Aug 2025

Software Engineering Co-op (Cloud Services and Integration)

Roy, UT

- Developed 10+ automated validation scripts in Jython and Groovy to scan thousands of UML system models for security and deployment errors. Reduced manual review time by 60% and prevented critical integration failures before release.
- Refactored and optimized a complex CI/CD pipeline, cutting deployment time from 8+ hours to under 10 minutes. Supported faster delivery cycles for 200+ internal projects.
- Built and maintained microservice components using enterprise patterns common in AWS and Java-based environments, improving system reliability and supporting seamless deployment across distributed teams.
- *AWS, Java, Jython, Groovy, C++, Git, CI/CD, Agile*

Northrop Grumman

May 2024 – Aug 2024

Software Engineering Intern (Physical Security)

Roy, UT

- Implemented back-end linear algebra algorithms in C to convert live incoming raw LIDAR data packets into structured point cloud (PCD) files, improving processing speed by 3x and delivering a smoother visualization for security and other developers.
- Developed front-end features in React to render interactive 3D point clouds, enabling non-technical users from 5+ internal teams to quickly analyze sensor data directly in a web application.
- *TypeScript, React, JavaScript, C, Node.js, REST APIs, Git, Linux, Jira, Agile*

University of Utah

Aug 2023 – May 2024

Teaching Assistant (Physics for Scientists and Engineers)

Salt Lake City, UT

- Led bi-weekly lab sessions and office hours for over 100 students, providing individualized support and fostering deeper engagement with core physics concepts throughout the semester.
- Designed and delivered over 20 interactive workshops and demonstrations that simplified complex topics like mechanics and electromagnetism, resulting in a 10% increase in final course grades among students.

Projects

- **Nuclixa — Summer 2026:** Developed an AI agent package registry and marketplace enabling developers to discover, publish, install, and manage AI agents, MCP servers, skills, and workflows. Architected scalable package management, search and discovery, metadata indexing, developer tooling, and secure distribution infrastructure to streamline reusable AI agent ecosystems and accelerate agent-based application development.
Next.js, React, TypeScript, Tailwind CSS, PostgreSQL, Prisma, OpenAI API, REST APIs
- **WikiSpeedrunning.com — Spring 2026:** Built a full-stack web platform for collaborative route optimization and competitive speedrunning of Wikipedia and other wiki articles. Implemented ranked matchmaking, Elo ratings, leaderboards, route tracking, and real-time multiplayer game state synchronization. Designed scalable backend services and optimized database queries to support persistent user profiles and competitive gameplay.
Next.js, TypeScript, PostgreSQL, Prisma, Redis, Supabase
- **Mechanical Detection of Magnetic Resonance — Undergraduate Research (2025 – 2026):** Designed and implemented software in Python and MATLAB to process, log, and visualize voltage signals from a custom magnetic-sensing device, achieving a 10x improvement in detection sensitivity from the initial design and demonstrating potential for next generation magnetic sensors and spin-detection technologies.
Python, MATLAB, Signal Processing